5-MINUTE REVISION

Measures of Dispersion

From the chapter's own revision points, formulae and questions.

Selected from Chapter 6 · Chapter Summary in the textbook.

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STAT 3202 · Chapter revision

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Quick revision · 1

- Dispersion → extent to which observations deviate from a central value.
- Range → difference between largest and smallest observations.
- Range → simplest measure of dispersion.
- IQR → spread of the middle 50% of observations.
- IQR → $Q_3 - Q_1$.

Quick revision · 2

- IQR is less affected by extreme values than range.
- MAD → average absolute deviation from a central value.
- MAD about the median is the least.
- Variance → average squared deviation, using $n - 1$ for sample variance.
- Standard deviation → non-negative square root of sample variance.

Quick revision · 3

- Variance is expressed in squared units.
- Standard deviation is expressed in the original units.
- Standard deviation is always non-negative.
- Standard deviation = 0 → all observations are identical.
- Larger standard deviation → greater dispersion, when datasets are measured on the same scale.

Quick revision · 4

- Standard deviation is an absolute measure of dispersion.
- Coefficient of variation is a relative measure of dispersion.
- Higher CV → greater relative variability.
- Lower CV → lower relative variability.
- CV is useful for comparing datasets measured in different units or scales.

Quick revision · 5

- Range uses only the two extreme observations.
- IQR uses the middle 50% of the observations.
- MAD uses all observations but ignores the signs of deviations.
- Standard deviation uses all observations and gives greater weight to extreme deviations because deviations are squared.

Important formulae · 1

Range:

$$R = L - S$$

where L is the largest observation and S is the smallest observation.

Interquartile range:

$$IQR = Q_3 - Q_1$$

Mean absolute deviation:

$$MAD = \frac{\sum_{i=1}^{n} |x_i - \bar{x}|}{n}$$

Important formulae · 2

Sample variance:

$$s^2 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n - 1}$$

Sample standard deviation:

$$s = \sqrt{\frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n - 1}}$$

Alternative formula for sample variance:

$$s^2 = \frac{1}{n-1} \left\{ \sum_{i=1}^{n} x_i^2 - \frac{1}{n} \left(\sum_{i=1}^{n} x_i \right)^2 \right\}$$

Important formulae · 3

Variance for a discrete frequency distribution:

$$s^2 = \frac{1}{n-1}\left\{\sum_{i=1}^{k} f_i x_i^2 - \frac{1}{n}\left(\sum_{i=1}^{k} f_i x_i\right)^2\right\}$$

Standard deviation for a discrete frequency distribution:

$$s = \sqrt{\frac{1}{n-1}\left\{\sum_{i=1}^{k} f_i x_i^2 - \frac{1}{n}\left(\sum_{i=1}^{k} f_i x_i\right)^2\right\}}$$

For grouped frequency data:

$$d_i = \frac{x_i - A}{c}$$

where x_i is the class mark, A is the class mark with the highest frequency, and c is the class interval.

Sample variance for grouped frequency data:

$$s^2 = \frac{1}{n-1}\left\{\sum_{i=1}^{k} f_i d_i^2 - \frac{1}{n}\left(\sum_{i=1}^{k} f_i d_i\right)^2\right\} c^2$$

Important formulae · 4

Sample standard deviation for grouped frequency data:

$$s = c\sqrt{\frac{1}{n-1}\left\{\sum_{i=1}^{k} f_i d_i^2 - \frac{1}{n}\left(\sum_{i=1}^{k} f_i d_i\right)^2\right\}}$$

Coefficient of variation:

$$CV = \frac{s}{\bar{x}} \times 100$$

Worked example from the chapter

Consider the measurement on yield and plant height of a paddy variety. The mean and standard deviation for yield are 50 kg and 10 kg respectively. The mean and standard deviation for plant height are 55 cm and 5 cm respectively. Compare the variability.

Solution

Here, the measurements for yield and plant height are in different units. Hence the variability can be compared only by using coefficient of variation.

For yield, CV = $\frac{10}{50} \times 100 = 20\%$

For plant height, CV = $\frac{5}{55} \times 100 = 9.1\%$

The yield is subject to more variation than the plant height.

Self-check from the chapter

1. The spread of observations is called _____. **[Dispersion]**
2. The difference between the largest and smallest observations is the _____. **[Range]**
3. The difference between the third and first quartiles is the _____ range. **[Interquartile]**
4. Half the interquartile range is called the quartile _____. **[Deviation]**